Simultaneous Equations & Inequalities

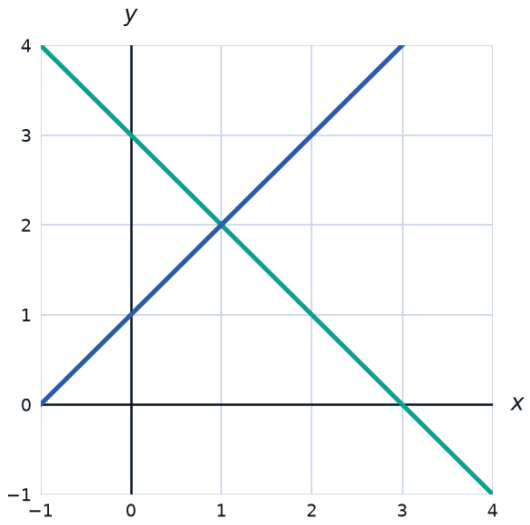
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Linear & Non-Linear Simultaneous Systems

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

2 [Foundational] • Target: Linear and Quadratic Inequalities

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.



3 [Foundational] • Target: Geometrical Applications & Line-Curve Systems

Solve the inequality $x^2 + 9x + 20 < 0$.

4 [Foundational] • Target: Advanced Problem Solving & Exam Challenges

Solve the inequality $x^2 + 4x + 3 < 0$.

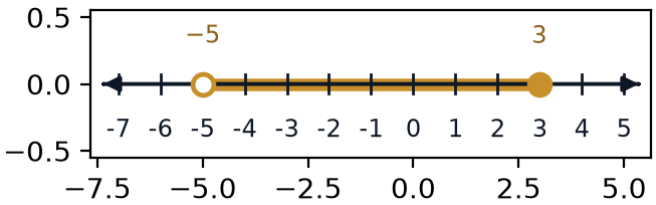
5 [Foundational] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 - x - 6 < 0$.

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6 [Foundational] • Target: Linear and Quadratic Inequalities

Solve the inequality $x^2 + 3x - 10 < 0$.



7 [Foundational] • Target: Geometrical Applications & Line-Curve Systems

Solve the inequality $x^2 + x - 20 < 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

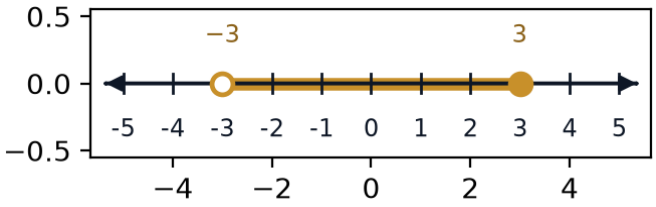
Solve the inequality $x^2 + 2x - 15 < 0$.

9 [Intermediate] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 - x - 56 < 0$.

10 [Intermediate] • Target: Linear and Quadratic Inequalities

Solve the inequality $x^2 + 4x - 21 < 0$.



11 [Intermediate] • Target: Geometrical Applications & Line-Curve Systems

Solve the inequality $x^2 + 6x - 7 < 0$.

12 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

Solve the inequality $x^2 + 5x - 36 < 0$.

13 [Intermediate] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 + 3x + 2 > 0$.

14 [Intermediate] • Target: Linear and Quadratic Inequalities

Solve the inequality $x^2 + 7x + 12 > 0$.

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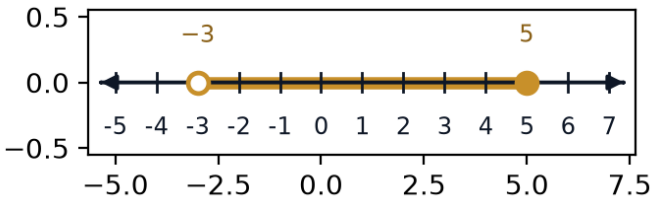
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Geometrical Applications & Line-Curve Systems • Cross-Lesson: 16.01 Permutations & Combinations

Solve the inequality $x^2 + 17x + 72 > 0$.

16 [Challenging] • Target: Advanced Problem Solving & Exam Challenges • Cross-Lesson: 17.01 Discrete Random Variables

Solve the inequality $x^2 - 10x + 21 < 0$.



17 [Challenging] • Target: Linear & Non-Linear Simultaneous Systems • Cross-Lesson: 18.01 Binomial Distribution

Solve the inequality $x^2 - 4x - 5 > 0$.

18 [Challenging] • Target: Linear and Quadratic Inequalities • Cross-Lesson: 19.01 Normal Distribution

Solve the inequality $x^2 - 6x - 7 > 0$.

19 [Challenging] • Target: Geometrical Applications & Line-Curve Systems • Cross-Lesson: 3.01 Graphs & Transformations

Solve the inequality $x^2 + x - 20 > 0$.

20 [Challenging] • Target: Advanced Problem Solving & Exam Challenges • Cross-Lesson: 4.01 Coordinate Geometry

Solve the inequality $x^2 + 7x - 8 > 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Linear & Non-Linear Simultaneous Systems

Solve the simultaneous equations $y = 2x - 3$ and $x^2 + y^2 = 26$. Which pair of points represents the solutions?

a. $(1, -1)$ and $\left(\frac{17}{5}, \frac{19}{5}\right)$

b. $(3, 3)$ and $(-1, -5)$

c. $(5, 7)$ and $(1, -1)$

d. $(2, 1)$ and $(-2, -7)$

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22 [Foundational] • Target: Linear & Non-Linear Simultaneous Systems

The line $x + y = 5$ intersects the hyperbola $xy = 6$. Find the coordinates of the two intersection points.

a. $(2, 3)$ and $(3, 2)$

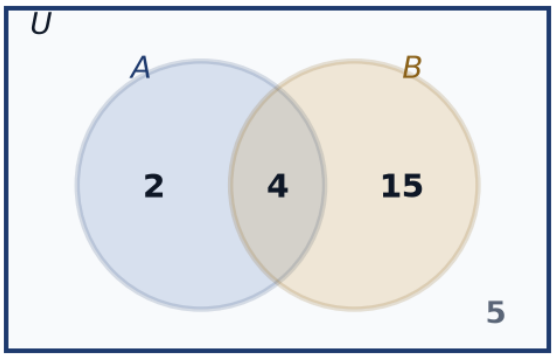
b. $(1, 6)$ and $(6, 1)$

c. $(2, 3)$ and $(-2, -3)$

d. $(4, 1)$ and $(1, 4)$

23 [Foundational] • Target: Linear & Non-Linear Simultaneous Systems

Find the points of intersection of the line $2x - y = 1$ and the curve $y = x^2 - 4$.



a. $(3, 5)$ and $(-1, -3)$

b. $(2, 3)$ and $(-2, -5)$

c. $(4, 7)$ and $(1, 1)$

d. $(5, 9)$ and $(0, -1)$

24 [Foundational] • Target: Linear and Quadratic Inequalities

Solve the quadratic inequality $x^2 - 2x - 8 < 0$.

a. $-2 < x < 4$

b. $x < -2$ or $x > 4$

c. $-4 < x < 2$

d. $x < -4$ or $x > 2$

25 [Foundational] • Target: Linear and Quadratic Inequalities

Find the set of values of x satisfying $2x^2 + 5x - 3 \geq 0$.

a. $x \leq -3$ or $x \geq \frac{1}{2}$

b. $-3 \leq x \leq \frac{1}{2}$

c. $x \leq -\frac{1}{2}$ or $x \geq 3$

d. $-1 \leq x \leq 3$

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26 [Foundational] • Target: Linear and Quadratic Inequalities

Solve the simultaneous inequalities $3x - 5 < 7$ and $x^2 - 9 \geq 0$.

a. $x \leq -3$ or $3 \leq x < 4$

b. $-3 \leq x \leq 3$

c. $x < 4$

d. $3 \leq x < 4$

27 [Foundational] • Target: Geometrical Applications & Line-Curve Systems

For what range of values of m does the line $y = mx$ not intersect the curve $y = x^2 + 9$?

a. $-6 < m < 6$

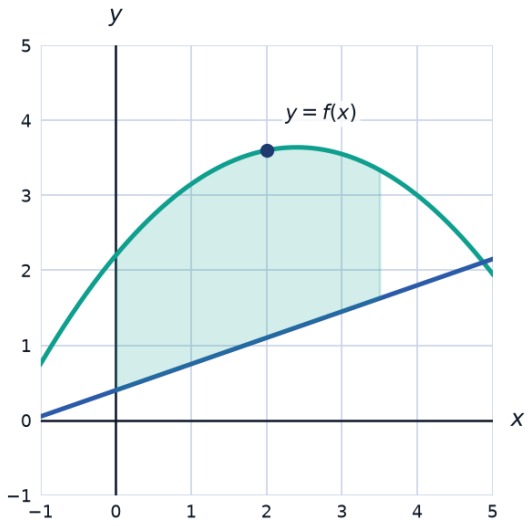
b. $m < -6$ or $m > 6$

c. $-3 < m < 3$

d. $m > 6$

28 [Foundational] • Target: Geometrical Applications & Line-Curve Systems

The perimeter of a rectangle is 28 cm and its area is at least 45 cm^2. What is the range of possible values for the width w ($w \leq l$)?



a. $5 \leq w \leq 7$

b. $4 \leq w \leq 9$

c. $3 \leq w \leq 7$

d. $w \geq 5$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Geometrical Applications & Line-Curve Systems

Find the coordinates of the points where the line $y = 3x - 2$ is normal to the curve $y = x^2 - 4x + 3$.

a. $(1, 0)$

b. $(2, -1)$

c. $(0, -2)$

d. $(3, 0)$

30 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

Find the range of values of k for which the line $y = kx - 1$ intersects the circle $x^2 + y^2 = 4$ in two distinct points.

a. $k^2 > -\frac{3}{4}$ (all real k)

b. $k > \sqrt{3}$

c. $k < -\sqrt{3}$ or $k > \sqrt{3}$

d. $-1 < k < 1$

31 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

A company finds that its profit P (in thousands of dollars) satisfies $P(x) = -2x^2 + 28x - 80 \geq 10$, where x is the production level in hundreds of units. What is the valid production interval for x?

a. $5 \leq x \leq 9$

b. $4 \leq x \leq 10$

c. $3 \leq x \leq 11$

d. $6 \leq x \leq 8$

32 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

Find all values of x satisfying the rational inequality $\frac{x-3}{x+2} < 0$.

a. $-2 < x < 3$

b. $x < -2$ or $x > 3$

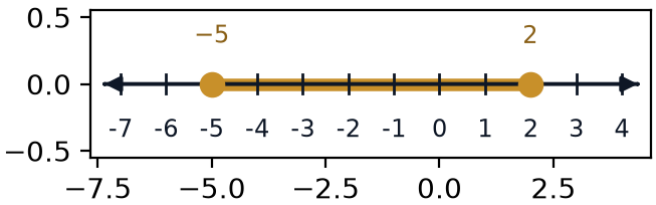
c. $x < 3$

d. $-3 < x < 2$

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33 [Intermediate] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 - 25 > 0$.



a. $x6$

b. $x5$

c. $x \leq -5$ or $x \geq 5$

d. -5<x<5

34 [Intermediate] • Target: Linear and Quadratic Inequalities

Solve the inequality $x^2 - x - 20 > 0$.

a. $x5$

b. $x \leq -4$ or $x \geq 5$

c. $x4$

d. -4<x<5

35 [Intermediate] • Target: Geometrical Applications & Line-Curve Systems

Solve the inequality $x^2 + 2x - 35 > 0$.

a. $x7$

b. -7<x<5

c. $x5$

d. $x \leq -7$ or $x \geq 5$

36 [Intermediate] • Target: Advanced Problem Solving & Exam Challenges

Solve the inequality $x^2 - 8x + 12 > 0$.

a. $x \leq 2$ or $x \geq 6$

b. 2<x<6

c. $x6$

d. $x - 2$

37 [Intermediate] • Target: Linear & Non-Linear Simultaneous Systems

Solve the inequality $x^2 - 9x + 20 > 0$.

a. $x5$

b. 4<x<5

c. $x - 4$

d. $x \leq 4$ or $x \geq 5$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Linear and Quadratic Inequalities • Cross-Lesson: 19.01 Normal Distribution

Solve the inequality $x^2 - 6x < 0$.

a. 0<x<6

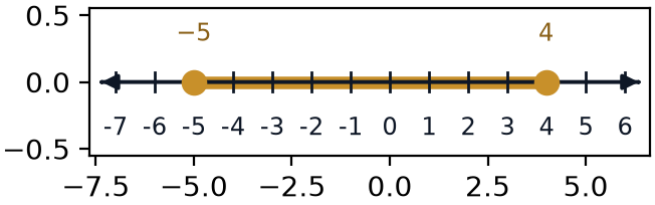
b. $0 \leq x \leq 6$

c. $x - 6$

d. $x6$

39 [Challenging] • Target: Geometrical Applications & Line-Curve Systems • Cross-Lesson: 3.01 Graphs & Transformations

Solve the inequality $x^2 - 3x < 0$.



a. $x - 3$

b. 0<x<3

c. $0 \leq x \leq 3$

d. $x3$

40 [Challenging] • Target: Advanced Problem Solving & Exam Challenges • Cross-Lesson: 4.01 Coordinate Geometry

Solve the inequality $x^2 + 6x > 0$.

a. -6<x<0

b. $x \leq -6$ or $x \geq 0$

c. $x6$

d. $x0$

41 [Challenging] • Target: Linear & Non-Linear Simultaneous Systems • Cross-Lesson: 5.01 Trigonometry

Find the x-coordinates of the points of intersection of $y = x^2 + 9x + 17$ and $y = -3$.

a. x=-5 or x=-5

b. x=-5 or x=-4

c. x=-4 or x=-4

d. x=5 or x=4

42 [Challenging] • Target: Linear and Quadratic Inequalities • Cross-Lesson: 6.01 Differentiation & Integration

Find the x-coordinates of the points of intersection of $y = x^2 + 10x + 9$ and $y = 2x - 3$.

a. x=-5 or x=-2

b. x=6 or x=2

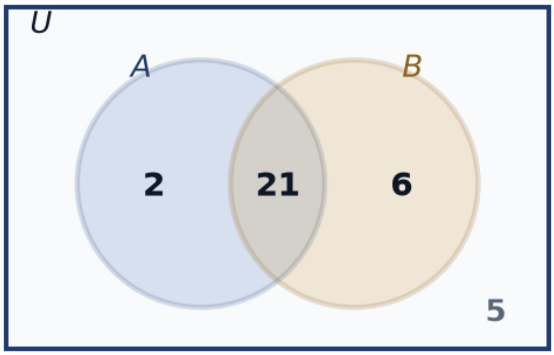
c. x=-6 or x=-2

d. x=-6 or x=-3

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43 [Challenging] • Target: Geometrical Applications & Line-Curve Systems • Cross-Lesson: 7.01 Statistics & Probability

Find the x-coordinates of the points of intersection of $y = x^2 - 2x - 21$ and $y = -6$.



a. x=-2 or x=5

b. x=3 or x=-5

c. x=-3 or x=4

d. x=-3 or x=5

44 [Challenging] • Target: Advanced Problem Solving & Exam Challenges • Cross-Lesson: 8.01 Functions: Composite & Inverse

Find the x-coordinates of the points of intersection of $y = x^2 + 4x + 8$ and $y = 5$.

a. x=-2 or x=-1

b. x=-3 or x=-1

c. x=-3 or x=-2

d. x=3 or x=1

45 [Challenging] • Target: Linear & Non-Linear Simultaneous Systems • Cross-Lesson: 9.01 Circular Measure

Find the x-coordinates of the points of intersection of $y = x^2 + 6x + 12$ and $y = 4$.

a. x=-3 or x=-2

b. x=-4 or x=-3

c. x=-4 or x=-2

d. x=4 or x=2

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